

MODULE 1

Introduction to Cybersecurity

Phase 1: Networking & Security Foundations | SSC Cybersecurity Training

Phase	Duration
Phase 1	Week 1

Module Overview

This foundational module introduces students to the cybersecurity field, establishing essential context for the entire 12-week program. Students will understand what cybersecurity professionals do, why their work matters, and what career paths exist in this rapidly growing field. This module serves as the gateway to the technical content that follows, ensuring all students begin with a shared understanding of core security principles and terminology.

Learning Objectives

By the end of this module, students will be able to:

1. Define cybersecurity and explain its importance in modern organizations
2. Identify and describe the CIA Triad (Confidentiality, Integrity, Availability)
3. Distinguish between different types of cyber threats and threat actors
4. Explain the basic components of the Cyber Kill Chain framework
5. Describe various cybersecurity career roles and their responsibilities
6. Identify the primary security domains (Network, Application, Data, Cloud, Physical)

Key Concepts & Terminology

What is Cybersecurity?

Cybersecurity is the practice of protecting computer systems, networks, programs, and data from digital attacks, unauthorized access, damage, or theft. These attacks typically aim to access, change, or destroy sensitive information, extort money from users, or interrupt normal business operations.

Why Cybersecurity Matters:

- **Financial Impact:** Global cybercrime damages are projected to reach \$10.5 trillion annually by 2025
- **Critical Infrastructure:** Attacks on power grids, water systems, and healthcare can endanger lives
- **Privacy Protection:** Personal data breaches affect millions of individuals annually
- **National Security:** State-sponsored cyber operations can impact elections, military operations, and diplomatic relations

The CIA Triad

The CIA Triad represents the three fundamental principles of information security. Every security control, policy, and procedure aims to protect one or more of these pillars:

1. Confidentiality

Ensuring that information is accessible only to authorized individuals. Confidentiality prevents unauthorized disclosure of sensitive data.

Examples:

- Encryption of data in transit and at rest
- Access controls requiring authentication (passwords, biometrics)
- Need-to-know policies limiting data access

2. Integrity

Maintaining the accuracy and completeness of data. Integrity ensures information is not altered in an unauthorized or undetected manner.

Examples:

- Hash functions to verify file integrity
- Digital signatures to authenticate data sources
- Version control systems tracking changes

3. Availability

Ensuring authorized users have reliable and timely access to information and resources when needed.

Examples:

- Redundant systems and backup servers
- DDoS protection to prevent denial-of-service attacks
- Regular system maintenance and updates

Cyber Threats & Threat Actors

Understanding who attacks systems and why is crucial for effective defense. Threat actors vary in motivation, sophistication, and resources.

Threat Actor	Motivation	Typical Targets
Script Kiddies	Curiosity, notoriety, proving skills	Poorly secured websites and networks
Cybercriminals	Financial gain through theft, fraud, ransomware	Financial institutions, healthcare, retail, individuals
Hacktivists	Political or social causes, ideological beliefs	Government agencies, corporations, controversial organizations
Nation-States	Military advantage, political disruption	Critical infrastructure, government agencies, defense contractors
Insider Threats	Revenge, financial gain, carelessness	Own employer, internal systems with privileged access

Common Attack Types

Attack Type	Description
Malware	Malicious software including viruses, worms, trojans, and ransomware designed to damage, disrupt, or gain unauthorized access to systems
Phishing	Fraudulent emails or messages impersonating trusted entities to trick users into revealing credentials or sensitive information
Denial of Service (DoS/DDoS)	Overwhelming a system with traffic to make it unavailable to legitimate users
Man-in-the-Middle (MitM)	Intercepting communications between two parties to eavesdrop or alter data in transit
SQL Injection	Inserting malicious SQL code into application queries to manipulate databases
Zero-Day Exploits	Attacks leveraging previously unknown vulnerabilities before patches are available

The Cyber Kill Chain

Developed by Lockheed Martin, the Cyber Kill Chain is a framework for understanding the stages of a cyberattack. By identifying which stage an attack is in, defenders can implement appropriate countermeasures.

- 1

Reconnaissance – Attackers research and gather information about the target
- 2

Weaponization – Creating malicious payloads (malware, exploit code)
- 3

Delivery – Transmitting the weapon to the target (email, USB, compromised website)
- 4

Exploitation – Triggering the vulnerability to execute malicious code
- 5

Installation – Installing malware or backdoors for persistent access
- 6

Command & Control (C2) – Establishing communications channel for remote control
- 7

Actions on Objectives – Achieving the attack goal (data theft, destruction, lateral movement)

Security Domains

Cybersecurity encompasses multiple specialized domains, each requiring distinct knowledge and skills:

Domain	Description
Network Security	Protecting network infrastructure using firewalls, intrusion detection/prevention systems, VPNs, and network segmentation
Application Security	Securing software through secure coding practices, input validation, security testing, and patch management
Data Security	Protecting sensitive information through encryption, data loss prevention (DLP), and access controls
Cloud Security	Securing cloud-based resources, managing shared responsibility models, and implementing cloud-native security controls
Endpoint Security	Protecting individual devices (laptops, phones, servers) with antivirus, EDR solutions, and hardening configurations

Physical Security

Controlling physical access to facilities, equipment, and infrastructure

Cybersecurity Career Pathways

The cybersecurity field offers diverse career paths with strong demand and competitive salaries. This program prepares students for entry-level defensive security roles, with clear pathways to specialization.

Entry-Level Roles (Target for This Program)

SOC Analyst (Tier 1) | Typical Salary: \$55,000 – \$75,000

Monitor security alerts, investigate incidents, and escalate threats

Security Analyst | Typical Salary: \$60,000 – \$80,000

Assess security posture, perform vulnerability scans, recommend improvements

Incident Responder (Junior) | Typical Salary: \$65,000 – \$85,000

Respond to security incidents, contain threats, assist with forensic analysis

Mid-Level Specializations

- Penetration Tester – Simulate attacks to find vulnerabilities before malicious actors do
- Threat Hunter – Proactively search for hidden threats within networks
- Security Engineer – Design and implement security systems and infrastructure
- Digital Forensics Analyst – Investigate cybercrimes and recover evidence from digital devices

Advanced & Leadership Roles

- Security Architect – Design enterprise security frameworks and strategies
 - SOC Manager – Lead security operations teams and incident response processes
 - Chief Information Security Officer (CISO) – Executive responsible for entire organizational security strategy
-

Additional Resources & Reading

Recommended Reading

- The Art of Invisibility by Kevin Mitnick – Practical privacy and security for everyday users
- Ghost in the Wires by Kevin Mitnick – True story of social engineering and hacking
- Countdown to Zero Day by Kim Zetter – The story of Stuxnet and nation-state cyber warfare

Online Resources

- NIST Cybersecurity Framework – <https://www.nist.gov/cyberframework>
- MITRE ATT&CK Framework – <https://attack.mitre.org>
- OWASP Top 10 – <https://owasp.org/www-project-top-ten/>
- US-CERT Alerts – <https://www.cisa.gov/uscert/ncas/alerts>

Industry News & Blogs

- Krebs on Security – In-depth cybersecurity journalism
 - The Hacker News – Daily cybersecurity news and analysis
 - Bleeping Computer – Technical security news and tutorials
-